



COLLEGE
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Orthopaedic conditions of the Hip joint

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Orthopaedic conditions of the Hip joint

Avascular necrosis of the hip (Hip osteonecrosis)

Hip Osteonecrosis, also known as avascular necrosis of the hip, represents a condition caused by reduced blood flow to the femoral head secondary to a variety of risk factors such as a traumatic event, sickle cell disease, steroid use, alcoholism, autoimmune disorders, and hypercoagulable states.



Epidemiology

Incidence

- 20,000 new cases per year in the United States
- accounts for 10% of total hip arthroplasties performed

Demographics

- male > females
- average age at presentation is 35 to 50

Anatomic location

- bilateral hips involved 80% of the time
- multifocal osteonecrosis
 - disease in three or more different joints
 - 3% of patients with osteonecrosis have multifocal involvement

Risk factors

direct causes

- irradiation
- trauma
- hematologic diseases (leukemia, lymphoma)
- dysbaric disorders (decompression sickness, "the bends") - Caisson disease
- marrow-replacing diseases (e.g. Gaucher's disease)
- sickle cell disease

indirect causes

- alcoholism
- hypercoagulable states
- steroids (either endogenous or exogenous)
- systemic lupus erythematosus (SLE)
- transplant patient
- virus (CMV, hepatitis, HIV, rubella, rubeola, varicella)
- protease inhibitors (type of HIV medication)
- idiopathic

Etiology

Pathophysiology

idiopathic AVN

- intravascular coagulation is the final common idiopathic pathway
- pathoanatomic cascade
 - coagulation of the intraosseous microcirculation →
 - venous thrombosis →
 - retrograde arterial occlusion →
 - intraosseous hypertension →
 - decreased blood flow to femoral head →
 - AVN of femoral head →
 - chondral fracture and collapse

AVN associated with trauma

- due to injury of femoral head blood supply (medial femoral circumflex)

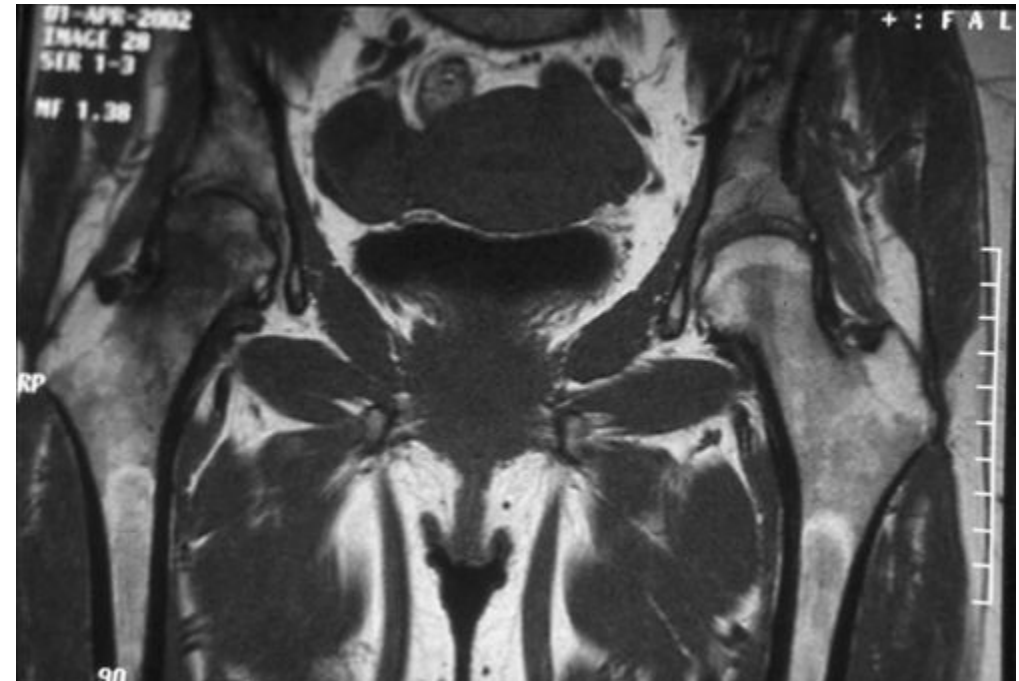
Classification

Steinberg Classification (modification of Ficat classification)			
Stage	Radiographs	MRI	
0	Normal	Normal MRI and bone scan	
I	Normal	Abnormal MRI and/or bone scan	
II	Cystic or sclerosis changes	Abnormal MRI and/or bone scan	
III	Crescent sign (subchondral collapse)	Abnormal MRI and/or bone scan	
IV	Flattening of femoral head	Abnormal MRI and/or bone scan	
V	Narrowing of joint	Abnormal MRI and/or bone scan	
VI	Advanced degenerative changes	Abnormal MRI and/or bone scan	

STAGE 0



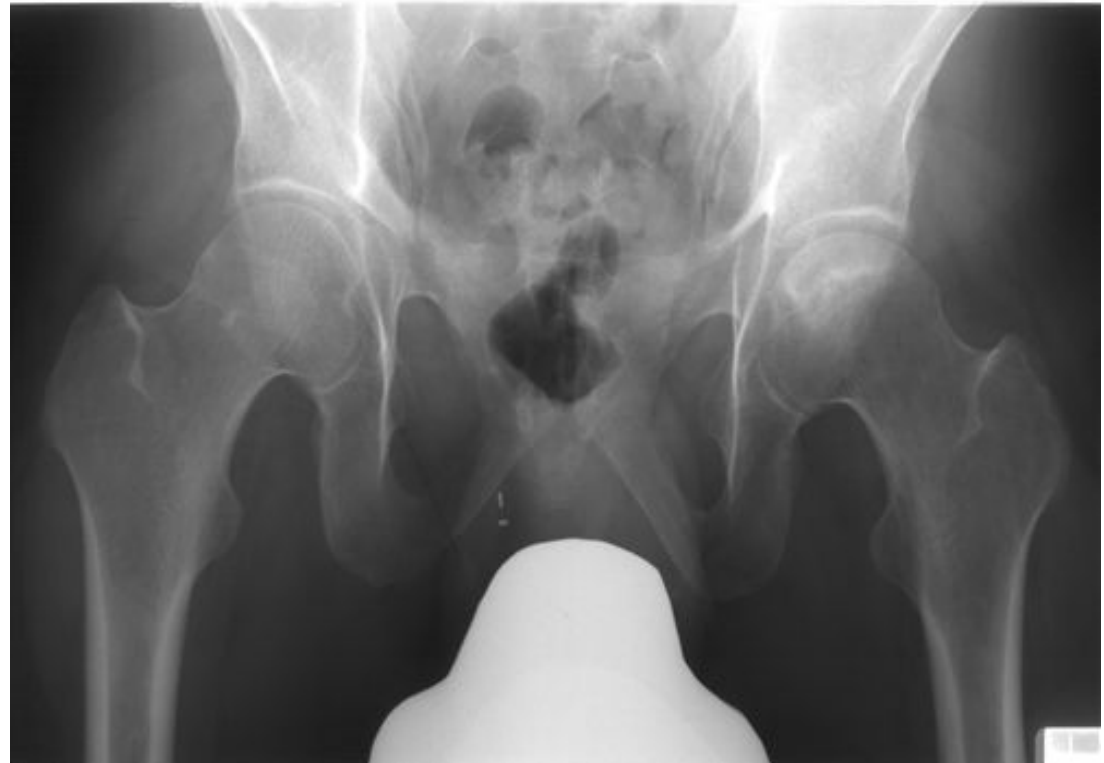
STAGE 1



STAGE 2



STAGE 3



STAGE 4



STAGE 5



Stage 6



Presentation

Presentation

Symptoms

- insidious onset of pain
- pain with stairs, inclines, and impact
- pain common in anterior hip

Physical exam

- mostly normal initially
- advanced stages similar to hip OA (limited motion, particularly internal rotation)

Imaging

- Radiographs

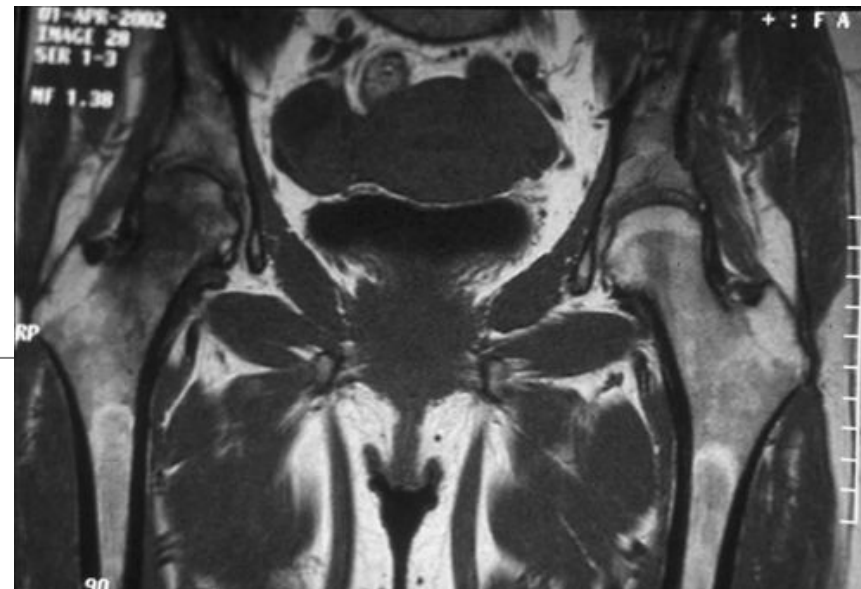
- recommended views

- AP hip
 - frog-lateral of hip
 - AP and lateral of contralateral hip
 - classification systems based largely on radiographic findings

- MRI

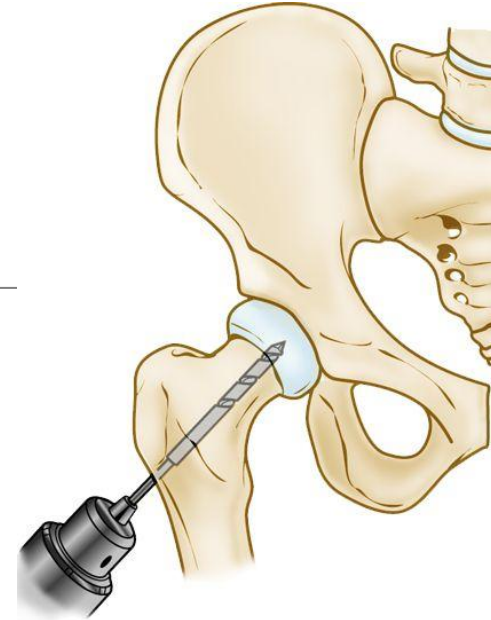
- highest sensitivity (99%) and specificity (99%)
 - double density appearance
 - T1: dark (low intensity band)
 - T2: focal brightness (marrow edema)
 - order when radiographs negative and osteonecrosis still suspected
 - presence of bone marrow edema on MRI is predictive of worsening pain and future progression of disease

- Bone scan

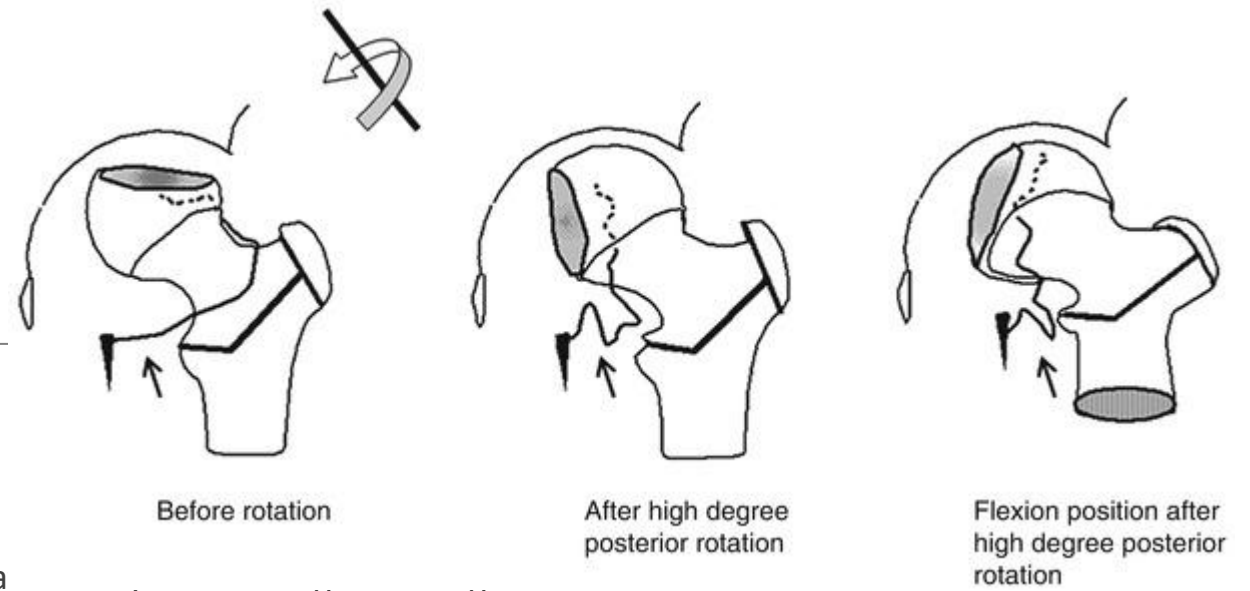


Treatment

- Nonoperative
 - **bisphosphonates**
 - indicated for precollapse AVN (Ficat stages 0-II)
- Operative
 - **core decompression with or without bone grafting**
 - indications
 - for early AVN, before subchondral collapse occurs
 - reversible etiology
 - technique
 - traditional method
 - drill an 8-10 mm hole through the subchondral necrosis
 - alternative method
 - pass a 3.2 mm pin into the lesion two to three times for decompression
 - relieves intraosseous hypertension equals less pain
 - stimulates a healing response via angiogenesis



Treatment



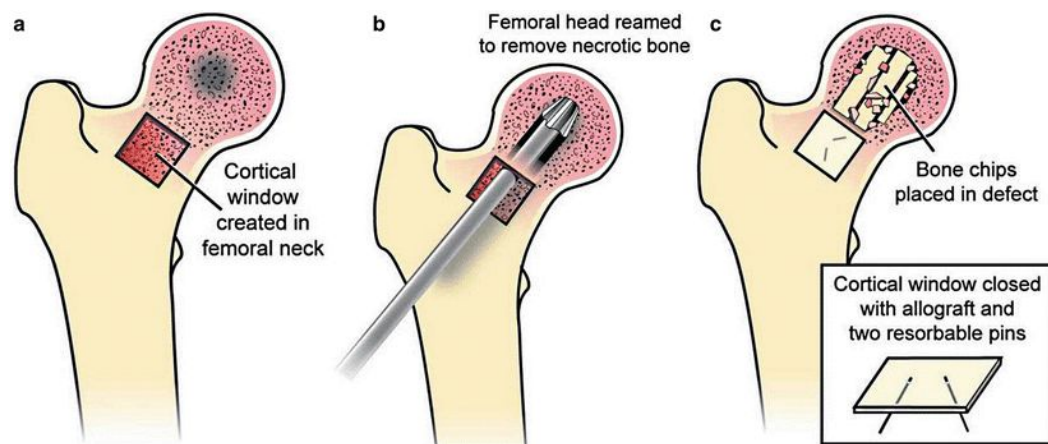
- **rotational osteotomy**

- indications
 - only for small lesions (<15%) in which the lesion can be rota

- **curettage and bone grafting through trapdoor technique or lightbulb technique**

- indications
 - preferably pre-collapse
- technique
 - lightbulb - through the cortex of the femoral neck-head junction to access the necrotic area of the femoral head and place bone graft
 - trapdoor - through articular surface

LIGHT BULB TECHNIQUE



TRAPDOOR TECHNIQUE

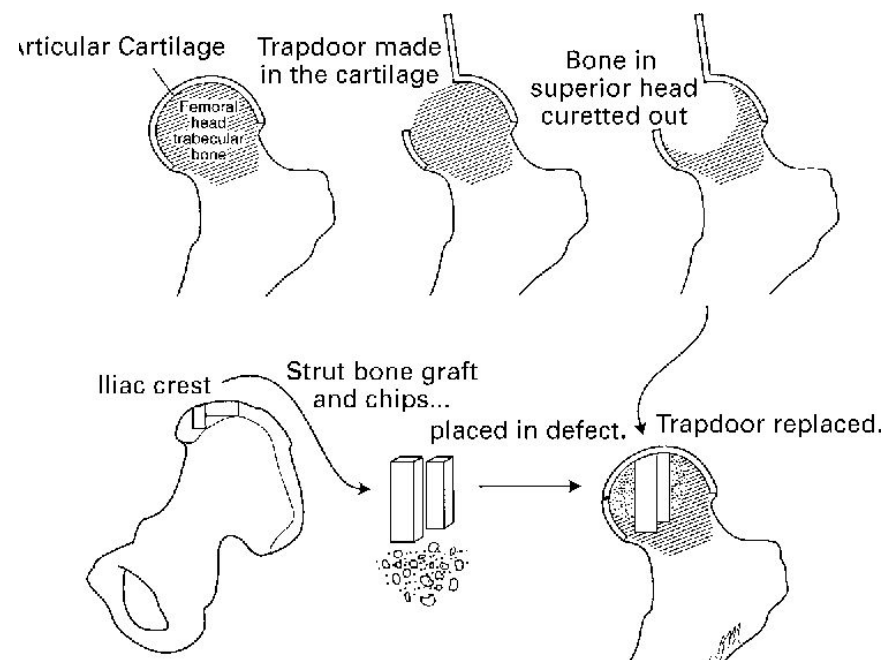


Fig. 1

Treatment

- **vascularized free-fibula transfer**
 - indications
 - for both pre-collapse and collapsed AVN in young patient
 - reversible etiology preferred
 - technique
 - remove the necrotic area with large core hole
 - fibular strut is placed under subchondral bone to help prevent collapse
 - complications
 - related to donor site morbidity
 - sensory deficit
 - motor weakness
 - FHL contracture
 - tibial stress fracture from side graft is taken

Treatment

- **total hip replacement**

- indications
 - younger patient with crescent sign or more advanced femoral head collapse, +/- acetabular DJD
 - irreversible etiology (chronic steroid use)
 - patients >40 with large lesions
- techniques
 - cementless cup and stem
- outcomes
 - in young patients with osteonecrosis, there is a higher rate of linear wear of the polyethylene liner and a higher rate of osteolysis than compared to older patients who have THA for osteoarthritis
 - provides good pain relief and function
 - most reliable means to provide pain relief and immediate return of function

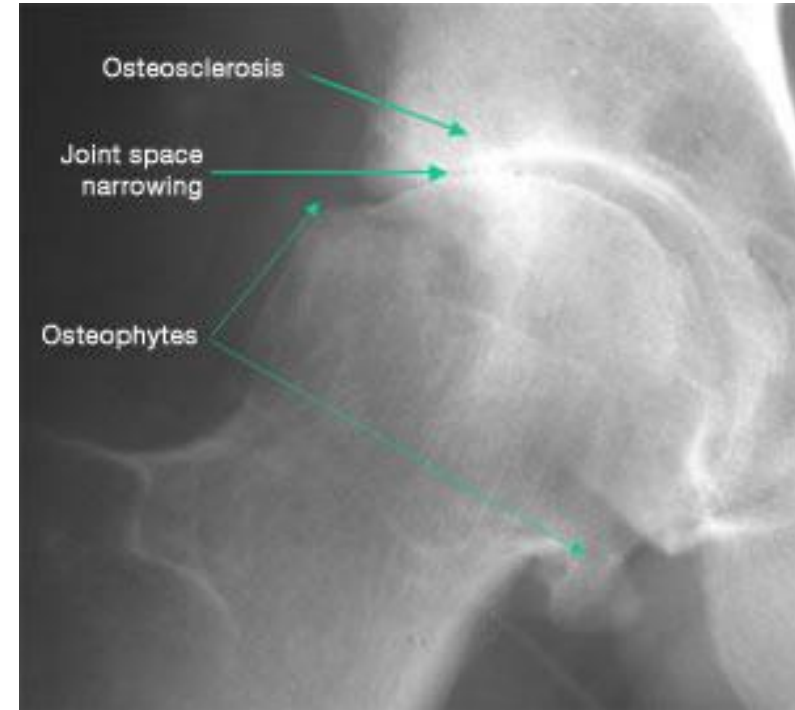


Treatment

- **total hip resurfacing**
 - indications
 - in advanced DJD with small, isolated focus of AVN
 - contraindicated in underlying disease process or chronic steroid use causing AVN (poor bone quality) and renal disease
- **hip arthrodesis**
 - indications
 - only consider in the very young patient in a labor intensive occupation

Hip Osteoarthritis

Hip Osteoarthritis is degenerative disease of the hip joint that causes progressive loss of articular cartilage of the femoral head and acetabulum.



Epidemiology

- Incidence
 - hip OA (symptomatic)
 - 88 per 100,000 per year
- Risk factors
 - modifiable
 - articular trauma
 - muscle weakness
 - heavy physical stress at work
 - high impact sporting activities
 - non-modifiable
 - gender
 - females > males
 - increased age
 - genetics
 - developmental or acquired deformities
 - hip dysplasia
 - slipped capital femoral epiphysis
 - Legg-Calvé-Perthes disease

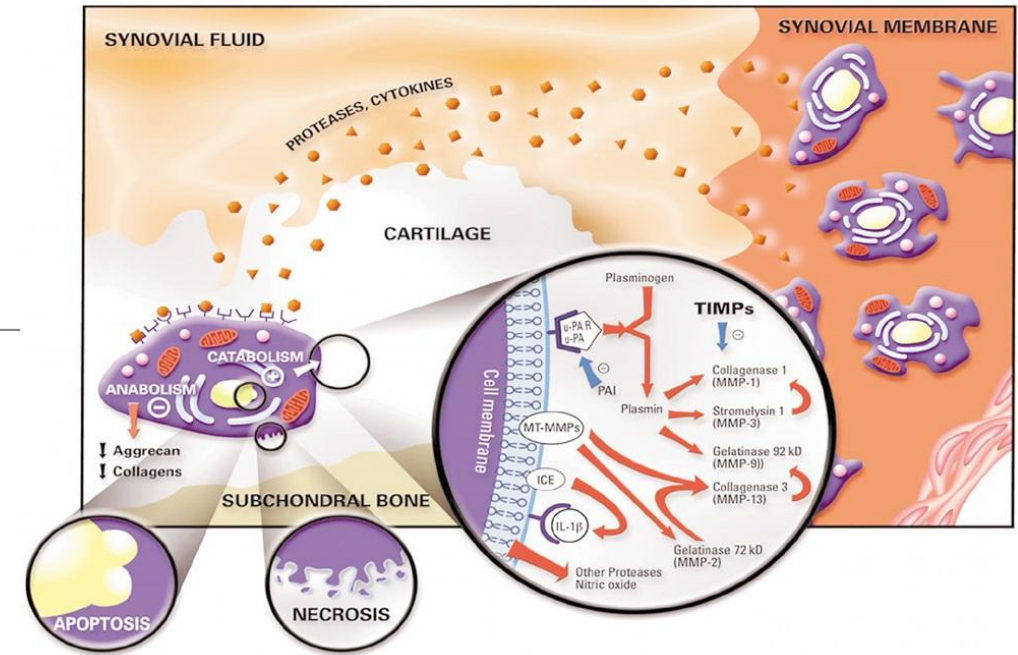
Etiology

- Pathophysiology

- pathoanatomy
 - articular cartilage
 - increased water content
 - decrease in amount of proteoglycans
 - Collagen organization and binding of proteoglycans to hyaluronic acid are lost
 - synovium and capsule
 - early phase of OA
 - mild inflammatory changes in synovium
 - middle phase of OA
 - moderate inflammatory changes of synovium
 - late phases of OA
 - synovium becomes increasingly thick and vascular
 - subchondral bone attempts to remodel
 - forming lytic lesion with sclerotic edges (different than bone cysts in RA)
 - bone cysts form in late stages

Etiology

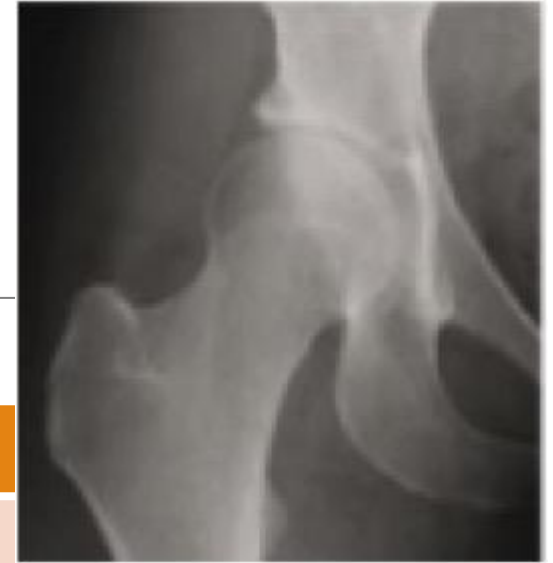
- Cell biology
 - proteolytic enzymes
 - matrix metalloproteases (MMPs)
 - responsible for cartilage matrix digestion
 - tissue inhibitors of MMPS (TIMPs)
 - control MMP activity preventing excessive degradation
 - imbalance between MMPs and TIMPs has been demonstrated in OA tissues
 - inflammatory cytokines
 - secreted by synoviocytes and increase MMP synthesis
 - examples
 - IL-1
 - IL-6
 - TNF-alpha



Etiology

- Genetics
 - genes potentially linked to OA
 - vitamin D receptor
 - estrogen receptor 1
 - inflammatory cytokines
 - IL-1
 - leads to catabolic effect

Classification



Tonnis grade 0

Tonnis Classification

Tonnis Classification		
Grade 0	• Normal radiographs	
Grade 1	• Sclerosis of femoral head and acetabulum • Slight joint space narrowing • Slight lipping at joint margins	
Grade 2	• Small cysts in femoral head/acetabulum • Moderate joint space narrowing • Moderate loss of head sphericity	
Grade 3	• Large cysts in femoral head/acetabulum • Joint space obliteration/severe narrowing • Severe femoral head deformity vs. AVN	

TONNIS GRADE 1



TONNIS GRADE 2



Tonnis grade 3



Presentation

- History

- identify age, functional activity, pattern of arthritic involvement, overall health and duration of symptoms

- Symptoms

- function-limiting groin pain
 - reduced walking distances
- pain at night or rest
- hip stiffness
- mechanical
 - instability, locking

-
- Physical exam
 - inspection
 - body habitus
 - gait
 - leg length discrepancy
 - skin (e.g. scars)
 - range of motion
 - lack of full extension (>5 degrees flexion contracture)
 - lack of full flexion (flexion < 90-100 degrees)
 - limited internal rotation
 - Neurovascular exam
 - straight leg test negative

Imaging

- Radiographs
 - recommended views
 - standing AP pelvis
 - AP + lateral hip
 - findings
 - osteoarthritis
 - joint space narrowing
 - osteophytes
 - subchondral sclerosis
 - subchondral cysts
 - pelvic obliquity
 - may be secondary to spinal deformity
 - may cause leg-length issues



Treatment

- Nonoperative

- **NSAIDs and/or tramadol**

- indications

- first line treatment for all patients with symptomatic arthritis

- **walking stick**

- decreases the joint reaction force on the affected hip when used in the contralateral upper extremity

- **weight loss, activity modification and exercise program/physical therapy**

- indications

- first line treatment for all patients with symptomatic arthritis

- BMI > 25

- **corticosteroid joint injections**

- indications

- can be therapeutic and/or diagnostic of symptomatic hip osteoarthritis

- **controversial treatments**

- viscoelastic joint injections

- glucosamine and chondroitin

52-year-old male with post traumatic AVN and secondary OA of the right hip (following fracture neck femur) treated with cementless THA , ceramic on polyethylene

PRE OP

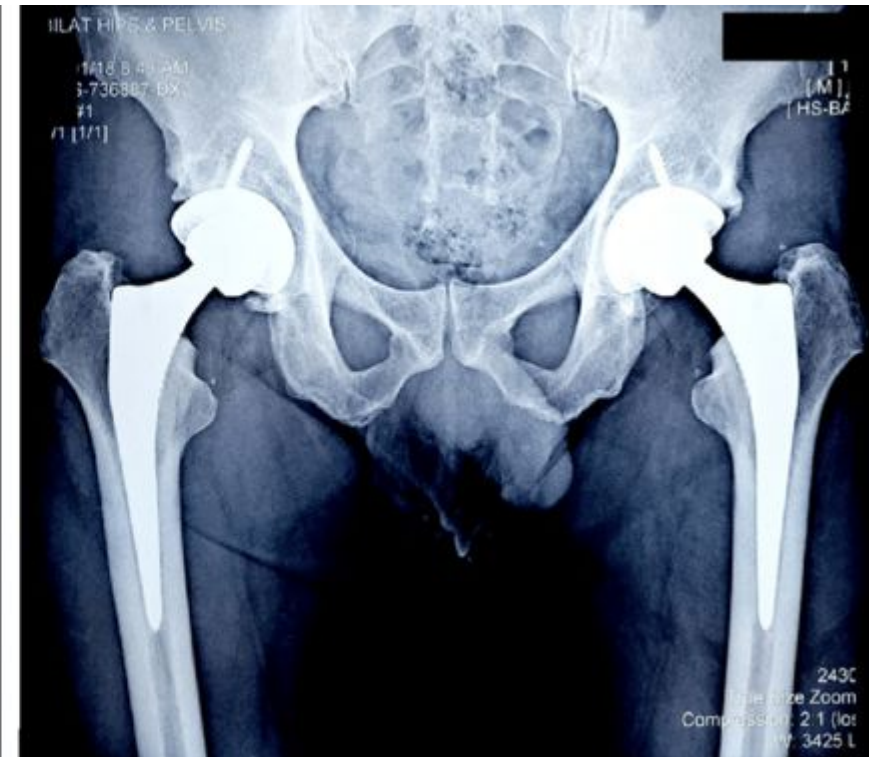


POST OP



Treatment

- Operative
 - **arthroscopic debridement**
 - indications
 - controversial, in labral tears
 - **periacetabular osteotomy +/- femoral osteotomy**
 - indications
 - symptomatic dysplasia in an adolescent or young adult with concentrically reduced hip and mild-to-moderate arthritis, can delay need for arthroplasty
 - **femoral head resection**
 - indications
 - pathological hip lesions
 - painful head subluxation
 - **hip resurfacing**
 - indications
 - young active, male, patients with hip osteoarthritis
 - **total hip arthroplasty (THA)**
 - indications
 - end-stage, symptomatic or severe osteoarthritis
 - preferred treatment for older patients (>50) and those with advanced structural changes

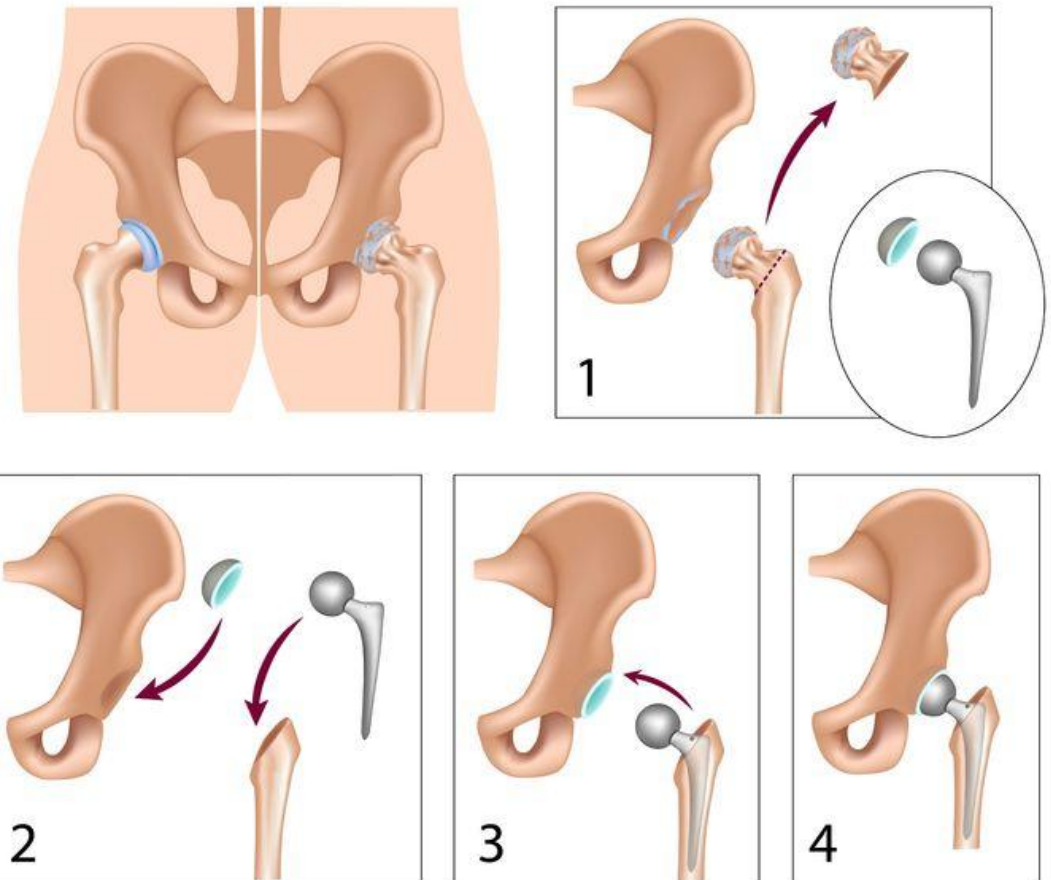


TOTAL HIP REPLACEMENT

Total replacement of the articular surfaces seems the ideal way of treating any disorder causing joint destruction. However, there are several problems to be overcome:

- (1) the prosthetic implants must be durable
- (2) they must permit extraordinary low friction movement at the articulation;
- (3) they must be firmly fixed to the skeleton.
- (4) they must be inert and not provoke any unwanted reaction in the tissues.

Total Hip Replacement



History

1891

- Dr. Gluck performs first reported attempt at a hip replacement with ivory used to replace the femoral head

1940

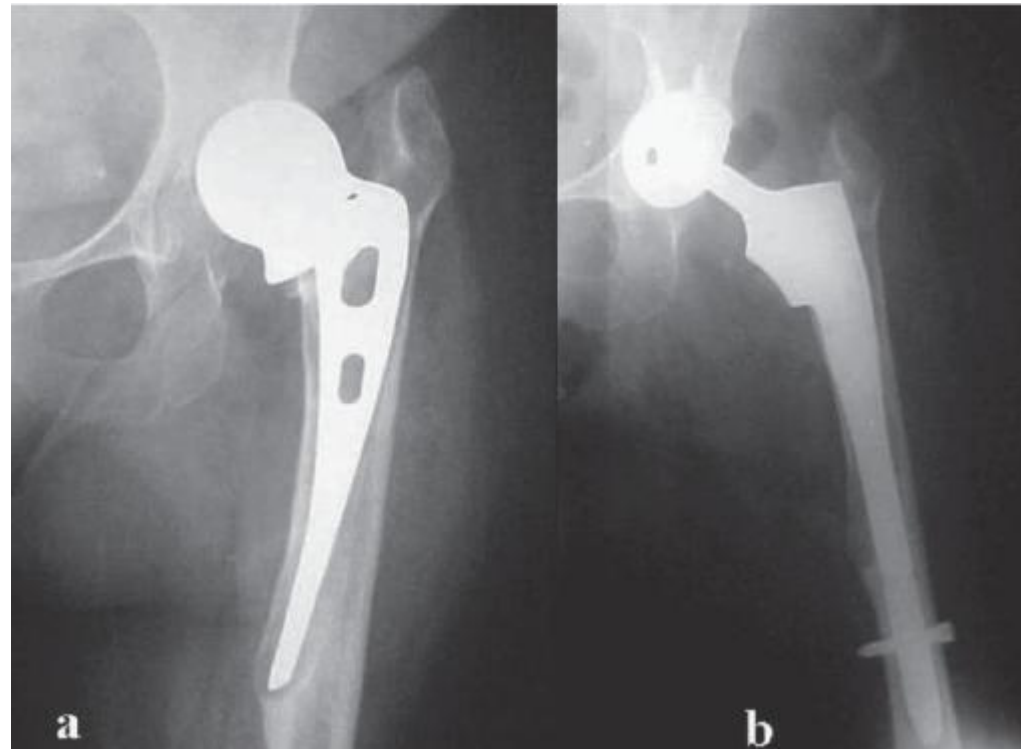
- Austin Moore performs first metallic hip replacement surgery (hemiarthroplasty) with a proximal femoral replacement bolted to the femur

1952

- Austin Moore prosthesis developed

1960s

- Sir John Charnley introduces concept of low friction arthroplasty concept
 - **termed "low friction"** as a small femoral head was used to reduce wear
 - components
 - metal femoral stem
 - polyethylene acetabular component
 - acrylic bone cement to fix the components



Using this implant, several authors have reported survivorship in the region of 80 per cent at a follow-up of 25 years. Total hip replacement reproducibly alleviates pain and restores mobility while providing joint stability. It has been described as 'the operation of the century'



INDICATIONS

The indications for hip replacement include pain, loss of movement and associated disability in the presence of radiographic evidence of joint destruction. Formerly, the procedure was largely restricted to the elderly and the infirm. The success of the early implants and vastly improved access to information have persuaded patients that an unacceptable compromise in the quality of life represents a valid indication for joint replacement. These patients expect to return to a full profile of professional and recreational activities. Given their increased expectations it is important that the risks and benefits of total hip replacement be fully discussed with them.

Contraindications

Overt or latent sepsis is the chief contraindication to joint replacement. An infected arthroplasty spells disaster. Patients under 60 years of age are considered if other operations are unsuitable.

Biomechanics of total hip arthroplasty depends

prosthesis design

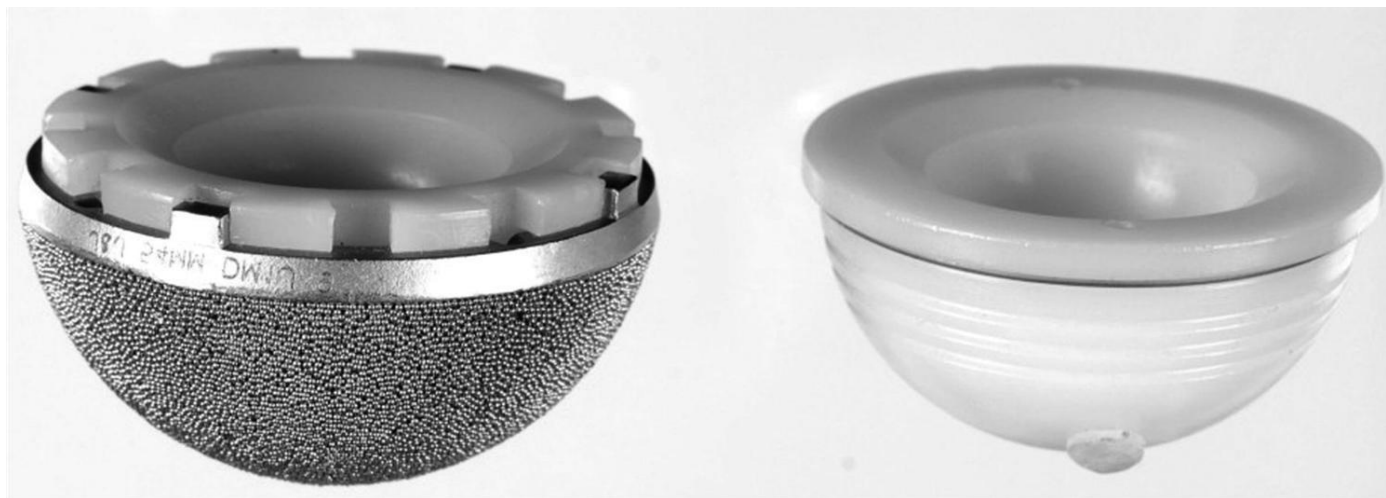
bearing surface and lubrication characteristics

fixation method

Designs include

- femoral component
 - cemented
 - press-fit (uncemented)
 - tapered stems
 - extensively porous coated stems
- acetabular components
 - cemented
 - polyethylene
 - metal
 - press-fit (uncemented)
 - metal





- bearing surfaces
 - polyethylene
 - metal
 - ceramic



Metal on Polyethylene



Metal on Metal



Ceramic on Ceramic

Press-fit femoral stems

Overview

- rely on biologic fixation
- compression hoop stresses provide initial stability

Cemented femoral stems

Overview

- rely on cement fixation
 - cement is a grout that provides initial and long-term stability
 - limited remodeling potential
 - preferred for irradiated bone due to the bone's limited ability for ingrowth
- composition
 - cobalt-chrome
 - most common
 - reduce cement stresses
 - titanium
 - may be prone to micromotion and debonding.



Bearing surfaces

Metal-on-polyethylene

- metal (cobalt-chrome) femoral head on polyethylene acetabular liner

Metal-on-metal

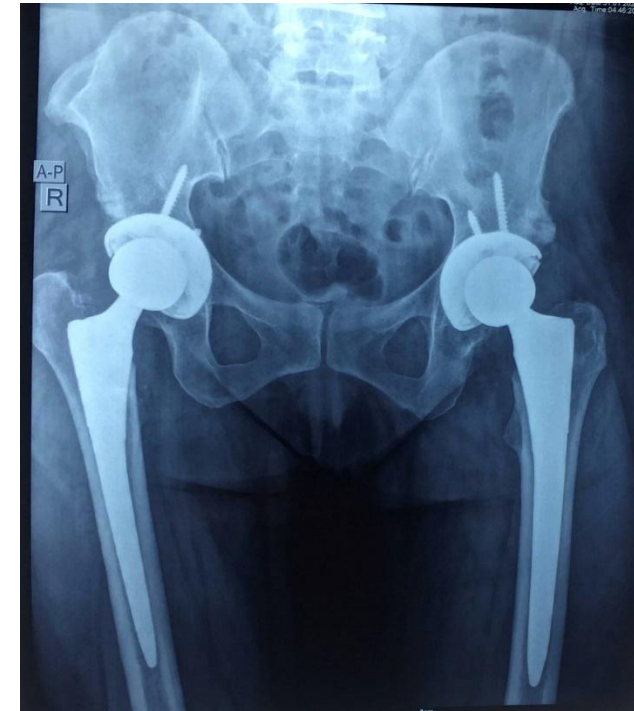
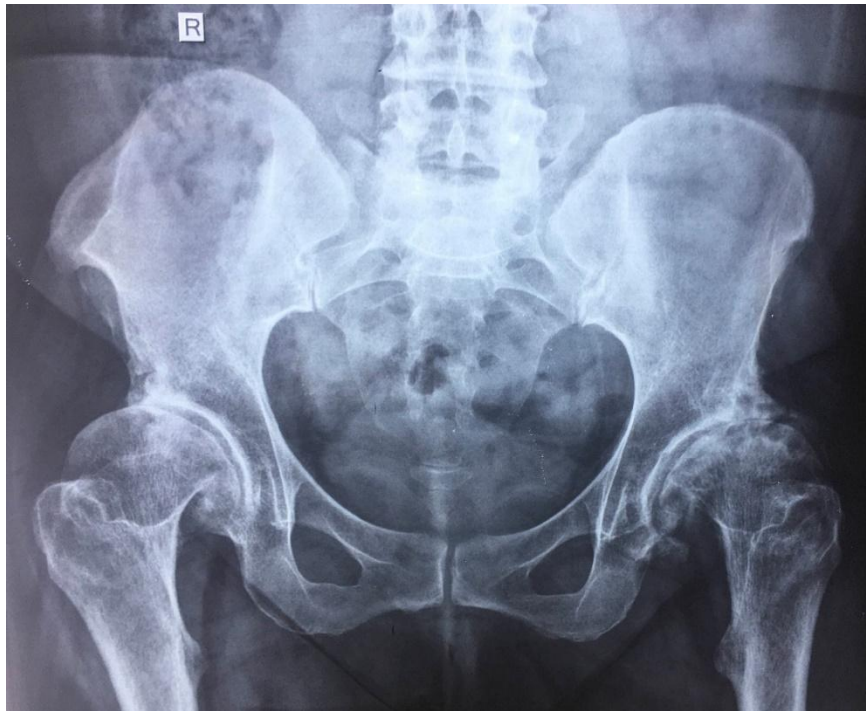
Ceramic on Ceramic

Ceramic on polyethylene

Titanium on Polyethylene



A 42 year old lady , with OA of both hip joints as an end result of DDH that had been treated with open reduction at an older age (8 years), finally treated successfully with simultaneous bilateral cementless THA .. Ceramic on poly ethylene bearing surface





Complications:

Acute:

Infection

THA dislocation

THA sciatic nerve palsy

THA vascular injury and bleeding

THA periprrothetic fracture

THA leg length discrepancy

Chronic:

THA aseptic loosening

Iliopsoas impingement

Heterotopic ossification

Infection

Thanks
